Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/27/2013
PI Name:	Naidu, Mamta Ph.D.		
Project Title:	High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02135	**Congressional District:**	7
Comments:	NOTE: Previous affiliation was State University of New York-Stony Brook, from October 2011-December 2012. Prior to that, affiliation was Brookhaven National Laboratory, Upton, NY, until September 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	09/15/2011	**End Date:**	06/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2013 (from 12/13/2013) due to PI move to new institution, per A. Chu/ARC (Ed., 1/29/2013) NOTE: Extended to 12/13/2013, per NSSC information (Ed., 12/19/12)		
Key Personnel Changes/Previous PI:	NOTE: New PI is Bruce Demple, per PI Naidu's move to GenSys Research Institute in January 2013 (Ed., 1/29/13) New Post Doctoral Fellow on this grant since April 2012, owing to move between institutions and SUNYSB procedures for transfer of H1; took 4 months to come through. At present post doc work on this grant has been at full salary April -Dec 2012 (with total 230K funding per year) and half salary Jan-June 2013 (due to 150K bridge money of which 27K (direct +indirect) was used for SUNYSB PDF salary.		
COI Name (Institution):	Demple, Bruce (SUNY-Stony Brook/ Adminstrative PI)		
Grant/Contract No.:	NNX11AO89G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note: New PI is Bruce Demple, per PI Naidu's move to GenSys Research Institute in January 2013 (Ed., 1/29/13). Grant is a continuation at SUNY-Stony Brook per PI's move in September 2011 from Brookhaven National Laboratory. See task, "High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo" (NNA10DA32I) for previous reports.] NASA's radiation research program is focused on assuring that humans can travel in space without exceeding the limiting risks from radiation, and on advancing the understanding of response mechanisms to radiation-caused DNA damage. Radiation studies on the CNS have predominantly focused on neurons, with few studies directly addressing the role of glial cells. The focus of our research proposal will be to determine the major DNA repair pathway(s) induced by HZE radiation in glial cells. Oligodendrocytes (OL) are responsible for forming and maintaining myelin, the insulating sheaths of neuronal axons. The damage to OL causes neurological impairment as OL have a slow turnover and are repopulated by resident OL progenitor cells (OPC). Astronauts may encounter repeated exposure to charged particles over the course of their missions such as traveling to Mars and this exposure to low doses of cosmic rays has the potential to prevent astronauts from successfully meeting mission requirements. Base excision repair(BER) is one of the predominant DNA damage repair pathways up regulated in response to oxidative stress by low LET radiation. Also, Apurinic endonuclease 1(Ape1), a key BER player may be involved in adaptive response by linking repair to oxidative pathways, thus, it becomes important to study its role in fractionated versus single low dose of high LET radiation. As Ape1 is known to affect the G1/S transition in hematopoietic embryoid bodies (EB) cells, it also becomes important to determine if cell cycle dependent changes in Ape1 levels can be correlated to differentiation capacity of rat glial progenitor cell CG-4 and in oligodendrocyte progenitor cells in thoracic region of rat spinal cord. Thus, our project will determine the effects of high LET radiation and protons on the nuclear and mitochondrial BER in two systems; the central glia-4 (CG-4), a glial progenitor cell line that can differentiate into OL or astrocytes, and in OPC cells in situ as well as those isolated from rat spinal cord. Also, we found neuroinflammation in these rats' brains which were not exposed to radiation. Finally, our recent work on cognitive function measurements from these rats reveals that Si ion at 300 MeV/n caused most Novel Object recognition test (NORT) defect when given as single low dose, at 1.5 months post exposure, where as Fe ion at 600 MeV/n and Protons at 250 MeV energy at same low dose were more detrimental 6-9 months post exposure, with fractionation resulting in higher defects in cognition with protons and Si, but Fe was most effective at single dose. The latent CNS effects are due to altered APE1 levels and altered OPC differentiation which may contribute to the observed neuro-inflammation and decreased cognitive function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study of the effects of fractionated, low dose versus single low dose of charged particles on glial cells of brain would help us understand the risk to astronauts exposed to chronic radiation exposures and the risk of normal tissues of patients receiving fractionated doses to their brain tumors. Our study will answer a critical question, "Would lower doses of higher Z particles produce more subtle latent changes, for example, changes in CNS cell population type that do not produce externally obvious deficits until months or years after exposure?" The oxidative stress produced by particles affects DNA repair capacity and our studies will show which DNA repair pathways are affected and how they affect the cells' differentiation responses to stress, thereby possibly altering cognitive behaviour, finally leading to better understanding of risks out there for our astronauts on long term missions.
Task Progress:	Aim 1. Will induction of APE1 be greater in rat brain derived CG-4 glial cells with fractionated, low doses of HZE radiation as compared to a single low dose of the same HZE radiation? Aim 2. Will fractionated, low dose HZE radiation modulate the differentiation capacity of glial progenitor cells (GPC) as compared to single low dose of HZE radiation? Results: 1) APE1 protein is marginally induced with X-rays but very significantly induced with 56Fe radiation, with similar localization profile in CG-4 cells. 2) Decreased mitochondrial membrane potential and increase in oxidative stress seen in response to 25-50 cGy of 56Fe, with fractionation versus single dose. 3) Our dose fractionation studies revealed a very significant (p value < 0.005) induction of total APE1 protein with fractionation as opposed to marginal induction with single doses and this increase correlated to number of fractions. 4) X-rays given as single dose of 1Gy inhibited mitochondrial membrane potential the most, as compared to fractionation of the same dose. 5) X-ray Fractionation induces higher immature oligodendrocyte and astrocyte formation. 6) Si and Fe beam (0.5 Gy dose) fractionation induce higher astrocyte formation only, as cells in OL differentiation medium senesce with in 4-6 days. 7) ROS increased with dose and fractionation for X-rays and Protons, but the 56Fe data were more variable. 8) Fractionated low LET radiation exposure of transiently suppressed APE1 CG-4 shows inhibition of APE1 induction, inhibition of cell differentiation resulting in premature cell death. Aim 3: Will fractionated, low dose of HZE radiation induce greater inhibition of OPC and OL populations in vivo? Results: 1)1 Gy of protons and 5 Gy of X-rays caused demyelination of rat spinal cord thoracic section after 1.5 month of exposure. 2) After 3 months of exposure to1 Gy of 250 MeV protons, we found increased proliferation of ependymal cells and destruction of the central canal, indicative of significant damage to spinal cords of these rats over time. 3) H & E staining of spinal cord sections isolated from 250 MeV protons (NSRL 11A) exposed rats after 3 months, showed significant destruction of cytoarchitecture of white matter and gray matter with clear demyelination as well as

	central canal destruction, more with fractionated dose than single dose of protons. 4) In vitro differentiation of OPCs purified from exposed rat spinal cords is a challenging experiment as the OPCs don't survive in culture beyond 2-4 days, so far under culture conditions used for these progenitor cells. 5) 1 Gy of X-rays appeared to be less damaging than 250 MeV protons after 9 months post exposure as measured by NORT cognition studies. 6) 0.5 Gy of 300 MeV/n 28Si is detrimental to NORT cognitive function in aged rats as compared to either 0.5 Gy of 600 MeV/n 56Fe or 1 Gy of 250 MeV Protons, 1.5 months after exposure. These rats are also more sluggish with bent spinal column than other exposed (Fe or protons) rats, so we have begun rotarod testing in all the NSRL animals to delineate between the cognitive defect and neuromuscular effect of HZE and proton radiation. After 6 months post exposure, Fe was most detrimental when given as single dose where as when given as fractionated dose (lower figure) both Si and Protons were more detrimental as compared to Fe. 7) Very significant Neuroinflammation with Proton and X-rays in aged rat brains after exposure to their thoracic spinal columns. 8) Non-targeted effects as tumor induction with dose fractionation of HZE and Protons over single dose. 9) 0.5 Gy of 600 MeV/n Fe induced most firing of neuron when given as single dose as compared to it being given in 2-3 fractions.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Suresh Kumar MA, Chaudhary P, Dhawan J, Biegon A, Kang E, Bennett R, Demple B, Naidu M. "Fractionated ionizing radiation skews differentiation of glial/ oligodendrocyte progenitor cells and induces cognitive defects." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. CNS Latent Effects to HZE and Protons forum; invited talk. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Articles in Peer-reviewed Journals	Demple B, Bennett R, Naidu M, Slatka R, Quinones J, Chen K, Thapar U, Suresh Kumar M. "DNA Maintenance for the Long Term: Base Excision Repair Systems for Endogenous DNA Damage in Mitochondria and the Nucleus." International Journal of Molecular Sciences. Invited review under submission as of August 2013. , Aug-2013